

10 Binomial Distributions

- Some probabilistic situations are so common that they have special names.
- In a sequence of **Bernoulli(p) trials**
 - There are only two possible outcomes, “success” (1) or not (0, “failure”), on each trial.
 - The unconditional/marginal probability of success is the same on every trial, and equal to p .
 - The trials are independent.
- There are several commonly used distributions associated with Bernoulli trials, including Geometric and Binomial.

Example 10.1. Consider an extremely simplified model for the daily closing price of a certain stock. Every day the price either goes up or goes down, and the movements are independent from day-to-day. Assume that the probability that the stock price goes up on any single day is 0.55. Let X be the number of days on which the price goes up in a 5-day week.

1. Compute and interpret $P(X = 0)$.
2. Compute the probability that the price goes up on the first day and then down on the following four days.
3. Why is $P(X = 1)$ different from the probability in the previous part? Compute and interpret $P(X = 1)$.

4. Compute and interpret $P(X = 2)$.
 5. Suggest a general formula for the probability mass function of X .
 6. Construct a table and plot to represent the distribution of X , and interpret it.
- Consider a *fixed number*, n , of Bernoulli(p) trials and let X count the number of successes. The distribution of X is defined to be the **Binomial(n, p) distribution**.
 - A Binomial distribution is specified by two parameters:
 - n (an integer): the fixed number of Bernoulli trials, or the sample size
 - p (in $[0, 1]$): the fixed marginal probability of success on each Bernoulli trial, or the population proportion of success
 - A discrete random variable X has a **Binomial distribution** with parameters n , a nonnegative integer, and $p \in [0, 1]$ if and only if its probability mass function is

$$p_X(x) = \binom{n}{x} p^x (1-p)^{n-x}, \quad x = 0, 1, 2, \dots, n$$

- In the Binomial(n, p) situation, there are $\binom{n}{x}$ S/F sequences that result in exactly x successes—and thus $n - x$ failures—in n trials,
- and each of these sequences has probability $p^x (1-p)^{n-x}$
- The *binomial coefficient* (read “ n choose x ”)

$$\binom{n}{x} = \frac{n!}{x!(n-x)!}$$

counts the number of success/failure sequences of length n in which there are exactly x successes. (Remember: by definition $0! = 1$.)

Example 10.2. Continuing Example 10.1.

1. Suggest a shortcut formula for $E(X)$.
 2. Compute $E(X)$ using the distribution of X from Example 10.1. Did the shortcut formula work?
 3. Interpret $E(X)$.
 4. Remember that we assumed $p = 0.55$. If instead p were 0.9, would $\text{Var}(X)$ be greater or less? What if $p = 0.5$?
- If X has a Binomial(n, p) distribution then

$$\begin{aligned}E(X) &= np \\ \text{Var}(X) &= np(1 - p)\end{aligned}$$

10.1 Exercises

Exercise 10.1. Continuing Example 10.1.

1. What does the random variable $5 - X$ represent? What is its distribution?

 2. Suppose that the price is currently \$100 and each it either moves up \$2 or down \$2. Let S be the stock price after 5 days. How does S relate to X ? Does S have a Binomial distribution?

 3. Recall that X is the number of days on which the price goes up in the next five days. Suppose that Y is the number of days on which the price goes up in the ten days after that (days 6-15). What is the distribution of Y ? What does $X + Y$ represent and what is its distribution? (Continue to assume independence between days, with probability 0.55 of an up movement on any day.)
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- If X and Y are independent, X has a $\text{Binomial}(n_X, p)$ distribution, and Y has a $\text{Binomial}(n_Y, p)$ distribution then $X + Y$ has a $\text{Binomial}(n_X + n_Y, p)$
 - A $\text{Binomial}(1, p)$ distribution is also known as a **Bernoulli**(p) distribution, taking a value of 1 with probability p and 0 with probability $1 - p$.
 - For a single trial, a Bernoulli random variable *indicates* if there is a success (1) or failure (0) on the trial
 - If $X_1, X_2, \dots, X_n$ are independent each with a $\text{Bernoulli}(p)$ distribution, then $X_1 + \dots + X_n$ has a $\text{Binomial}(n, p)$ distribution.
 - The random variable $X_1 + X_2 + \dots + X_n$ incrementally counts the total number of successes in the n trials

Exercise 10.2. In each of the following situations determine whether or not X has a Binomial distribution. If so, specify n and p . If not, explain why not.

1. Roll a die 20 times; X is the number of times the die lands on an even number.

2. Roll a die 20 times; X is the number of times the die lands on 6.
3. Roll a die until it lands on 6; X is the total number of rolls.
4. Roll a die 20 times; X is the sum of the numbers rolled.
5. Shuffle a standard deck of 52 cards (13 hearts, 39 other cards) and deal 5 *without replacement*; X is the number of hearts dealt. (Hint: be careful about why.)
6. Roll a fair six-sided die 10 times and a fair four-sided die 10 times; X is the number of 3s rolled (out of 20).
7. Randomly select a sample of 35 Cal Poly students; X is the number of students in the sample who are CA residents.

- Consider selecting a random sample of size n from a population in which every individual is classified as “success” or “failure”. Suppose that the proportion of individuals in the population classified as success is p .
- The individuals selected for the sample can be considered a sequence of Bernoulli(p) trials.
 - There are only two possible outcomes, success (1) and failure (0), on each trial.
 - The unconditional/marginal probability of success is the same on every trial, and equal to the population proportion of success p
 - The trials are independent
 - * If sampling with replacement, or
 - * If sampling without replacement but the population size is much larger than the sample size n (which is usually the case in practice)
- If X counts the number of successes in a random sample of size n from a large population with population proportion of success p then X has a Binomial(n, p) distribution.

Exercise 10.3. Donny Dont is thoroughly confused about the distinction between a random variable and its distribution. Help him understand by providing a simple concrete example of *two different random variables* X and Y that have the *same distribution*. Can you think of X and Y that have the same distribution but for which $P(X = Y) = 0$?